

(Liquid + liquid) equilibria of (water + ethanol + cyclohexane) at the temperatures 298.15 K and 323.15 K

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Experimental tie lines for (water + ethanol + cyclohexane) have been determined at the temperatures 298.15 K and 323.15 K. An elevation of temperature led to the results: the immiscibility region contracts slightly; the distribution curves of ethanol shift in the direction of enhancing the solubility of ethanol in the organic phase; the plait-point shifts towards the aqueous phase at higher mole fraction of ethanol; and a selectivity in extraction increases extremely in the region of lower mole fraction of ethanol in the organic phase. The present results are also compared with those for the ternary mixtures with other C₆ hydrocarbons.

1. Introduction

In a previous study,⁽¹⁾ we measured the high-pressure (liquid + liquid) equilibria for three (water + ethanol + a C₈-alkanol) mixtures at the temperature 298.15 K and elucidated the effect of molecular structure on the abilities of the C₈ alkanols as extractants; 2-ethylhexan-1-ol was found to be the most suitable for extracting ethanol among the studied alkanols. To obtain more detailed information on the solvent ability in extracting ethanol from its aqueous mixtures, it is helpful to see how the tie lines for the relevant ternary mixtures depend on differences of molecular shape and structure in the extractants.

The purpose of this study was to determine precise tie lines for (water + ethanol + cyclohexane) at the temperatures 298.15 K and 323.15 K and to compare the results with those for ternary mixtures containing other C₆ hydrocarbons.^(2, 3) All measurements have been made at the pressure 0.1 MPa because of freezing of cyclohexane by compression. The tie-line results obtained are reported here and compared with available results^(4, 5) for the present mixture.

2. Experimental

Spectrograde cyclohexane from Dojin Chemical Co. was used without further purification; its refractive index measured at 298.15 K agreed to within 0.00005 with

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the literature value.⁽⁶⁾ All other materials were the same as those in the previous study.⁽¹⁾ The apparatus and experimental procedure were identical to those used previously.⁽²⁾ The sample mixtures were prepared by mass in a manner similar to that used before.⁽¹⁾ After equilibrium was reached, the mixtures taken from conjugate phases were analysed by g.c. using acetone as an internal standard; a (0.32 cm $\times$ 2 m) glass column packed with PEG-6000 Shimalite TPA was used at 353 K and a helium flow rate of 0.33 cm³ $\cdot$ s⁻¹. The compositions of the mixtures were determined with the aid of the calibrating curve from the peak-area of each component; the mean error of the compositions determined was within mass fraction ± 0.001 of any component.

3. Results and discussion

Experimental tie lines for the present mixture at the temperatures 298.15 K and 323.15 K are listed in table 1 and plotted in figure 1, where x_i is the mole fraction of component i and the superscripts ' and ' ' denote the aqueous and organic phases at equilibrium, respectively. Our measurements at 298.15 K are compared in figure 1(a) with the solubilities and tie lines reported by other workers,^(4,5) but no tie lines for others' values are there drawn to avoid complication; there is good agreement between the two sets of results. The values for (water + ethanol + *n*-hexane)⁽²⁾ and (water + ethanol + benzene)⁽³⁾ at 298.15 K are also plotted in the figure to see the influence of differences of molecular shape and structure in the C₆ hydrocarbons.

TABLE 1. Experimental tie-line results for $\{x_1\text{H}_2\text{O} + x_2\text{C}_2\text{H}_5\text{OH} + (1-x_1-x_2)\text{C}_6\text{H}_{12}\}$ at the temperatures 298.15 K and 323.15 K

Aqueous phase		Organic phase		Aqueous phase		Organic phase	
x'_1	x'_2	x''_1	x''_2	x'_1	x'_2	x''_1	x''_2
<i>T</i> = 298.15 K							
1	0	0	0	0.429	0.531	0.005	0.063
0.936	0.064	0.011	0.002	0.323	0.598	0.011	0.091
0.875	0.125	0.009	0.007	0.210	0.625	0.019	0.157
0.836	0.164	0.014	0.011	0.173	0.622	0.022	0.195
0.774	0.225	0.008	0.017	0.161	0.613	0.025	0.211
0.723	0.274	0.010	0.026	0.150	0.591	0.035	0.247
0.663	0.332	0.010	0.027	0.120	0.560	0.038	0.293
0.570	0.418	0.009	0.038	0.103	0.518	0.053	0.346
0.485	0.489	0.005	0.051	0.094	0.503	0.057	0.362
<i>T</i> = 323.15 K							
1	0	0	0	0.485	0.478	0.012	0.089
0.870	0.130	0	0.018	0.373	0.547	0.014	0.125
0.817	0.182	0.004	0.026	0.363	0.552	0.016	0.130
0.730	0.267	0.002	0.038	0.307	0.578	0.021	0.158
0.673	0.321	0.002	0.048	0.288	0.589	0.031	0.187
0.627	0.365	0.003	0.057	0.273	0.598	0.040	0.230
0.596	0.392	0.003	0.062	0.239	0.587	0.057	0.281
0.551	0.428	0.004	0.073	0.172	0.553	0.082	0.380
0.530	0.448	0.005	0.076	0.148	0.529	0.094	0.423

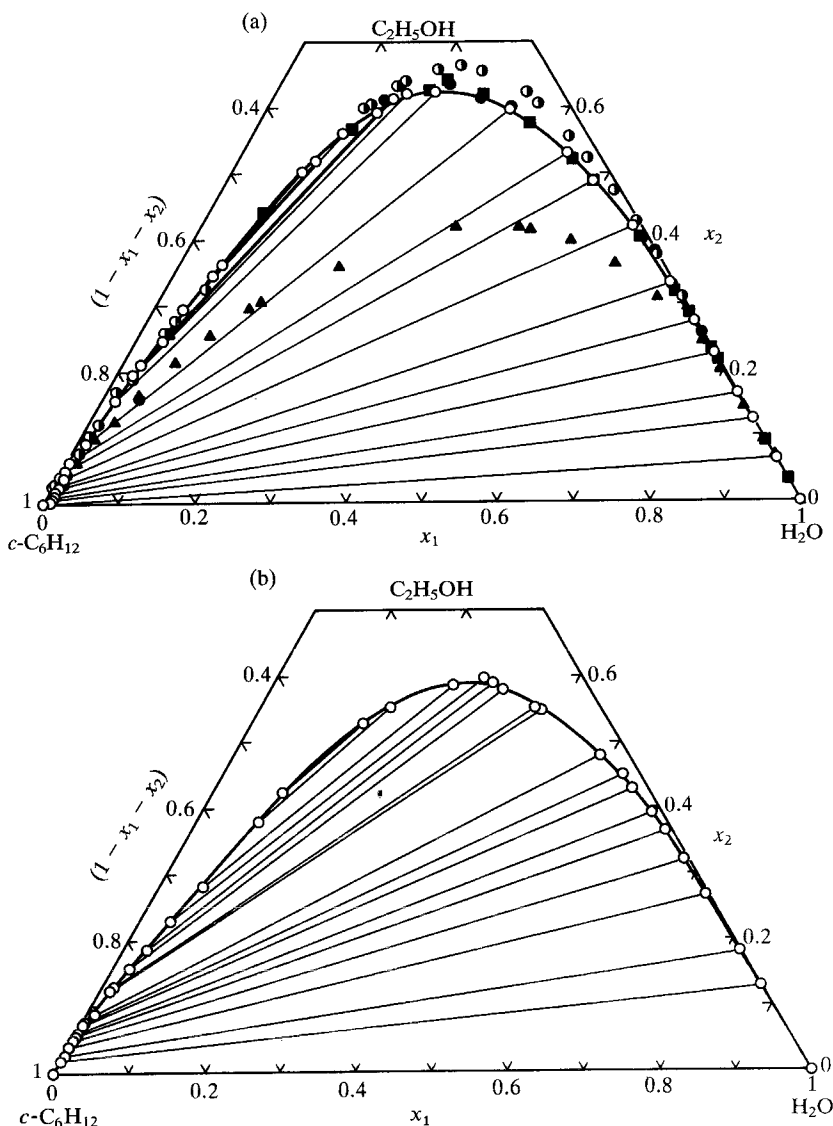


FIGURE 1. (Liquid + liquid) equilibria for $\{x_1\text{H}_2\text{O} + x_2\text{C}_2\text{H}_5\text{OH} + (1-x_1-x_2)\text{C}_6\text{H}_{12}\}$ at 0.1 MPa. (a), at 298.15 K: ○, this work; ■, Vold and Washburn;⁽⁴⁾ ●, Tarasenkova and Paulsen;⁽⁵⁾ ○, previous work⁽²⁾ for $\{x_1\text{H}_2\text{O} + x_2\text{C}_2\text{H}_5\text{OH} + (1-x_1-x_2)\text{C}_6\text{H}_{14}\}$; ▲, Yi-Chung and Moulton⁽³⁾ for $\{x_1\text{H}_2\text{O} + x_2\text{C}_2\text{H}_5\text{OH} + (1-x_1-x_2)\text{C}_6\text{H}_6\}$ and (b), at 323.15 K.

The solubilities for the present mixture are similar to those for (water + ethanol + *n*-hexane), except for the range of higher mole fractions of $\text{C}_2\text{H}_5\text{OH}$ in the aqueous phase; the immiscibility region reduces in the sequence: *n*-hexane, cyclohexane, and benzene, in agreement with the increasing magnitude of the Hildebrand solubility parameters⁽⁷⁾ for the hydrocarbons at 298.15 K. A comparison of the results for the

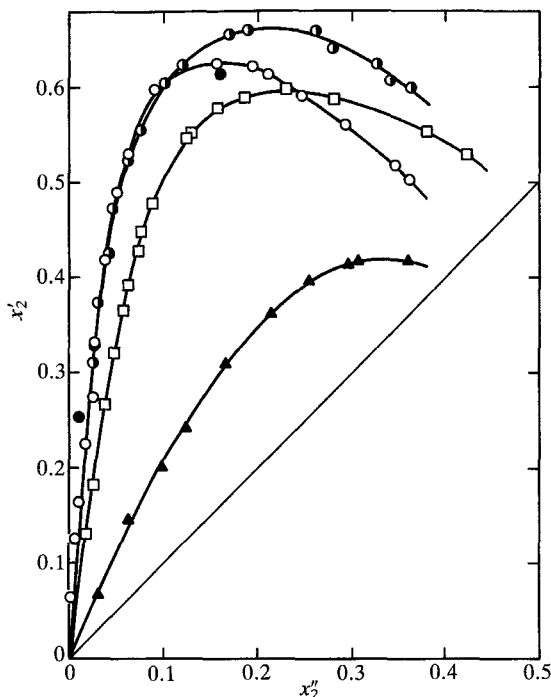


FIGURE 2. Distribution of ethanol between water and C_6 hydrocarbon. ○, and ● (Tarasenkova and Paul'sen),⁽⁵⁾ cyclohexane (298.15 K); □, cyclohexane (323.15 K); ●, n -hexane (298.15 K),⁽²⁾ ▲, benzene (298.15 K).⁽³⁾

present mixture in figures 1(a) and 1(b) shows that an elevation of temperature leads to a slight contraction of the immiscibility region and to a reduction of the tie-line slopes.

The distribution curves of ethanol at 298.15 K and 323.15 K are shown in figure 2 together with those of two other ternary mixtures at 298.15 K. The distribution curves deviate greatly above the reference line of slope unity, indicating preferential distribution of ethanol into the aqueous phase, and are located near the curve for (water + ethanol + n -hexane), far removed from that for (water + ethanol + benzene). The deviation of the curves becomes smaller at elevated temperatures, and the plait-point is expected to shift towards the aqueous phase at higher mole fraction of C_2H_5OH . The plait-point estimated as before⁽⁸⁾ from the measured values near the point is: $x_1 = (0.068 \pm 0.003)$ and $x_2 = (0.424 \pm 0.001)$ at 298.15 K, and $x_1 = (0.114 \pm 0.004)$ and $x_2 = (0.476 \pm 0.002)$ at 323.15 K.

The distribution coefficients D_2 of ethanol, defined as the ratio of the mole fraction x'_2 of C_2H_5OH in the aqueous phase to that x''_2 in the organic phase at equilibrium, are plotted against x''_2 in figure 3, together with the values obtained from available values for the present and two other mixtures. The values of Tarasenkova and Paul'sen⁽⁵⁾ for the present mixture at 298.15 K are in good agreement with ours and, further, those for (water + ethanol + n -hexane)⁽²⁾ lie virtually on the smooth

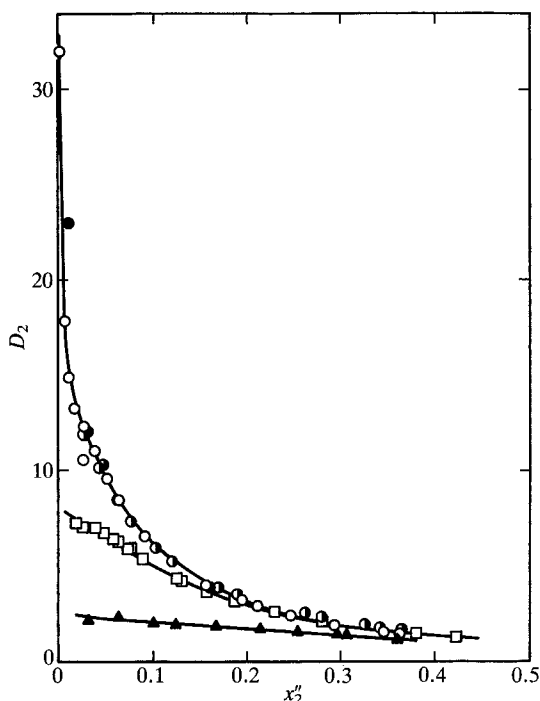


FIGURE 3. Distribution coefficients D_2 for ethanol as a function of mole fraction x_2'' of C_2H_5OH in the organic phase. ○, and ● (Tarasenko and Paul'sen),⁽⁵⁾ cyclohexane (298.15 K); □, cyclohexane (323.15 K); ●, *n*-hexane (298.15 K),⁽²⁾ ▲, benzene (298.15 K).⁽³⁾

curve drawn through our points. Elevation of temperature reduces D_2 remarkably in the region of lower x_2'' ; the D_2 curve at 323.15 K is, on the whole, gentler than that at 298.15 K; therefore, it can be noted that the solvent capacity of cyclohexane to extract ethanol from dilute aqueous mixtures becomes more favourable as the temperature rises. The D_2 values for (water + ethanol + benzene)⁽³⁾ are the smallest among the three mixtures compared and depend only a little on x_2'' ; the observations show that benzene is most excellent in solvent capacity.

The separation factor α , expressing a selectivity in extraction, is defined in terms of the distribution coefficient D_1 for water and that D_2 for ethanol by

$$\alpha = D_1/D_2, \quad (1)$$

where D_1 is given as the ratio x_1'/x_1'' of the mole fractions of H_2O in the two phases at equilibrium in a similar manner as D_2 . The α values at two temperatures calculated from the tie lines in table 1 are plotted against x_2'' in figure 4, where the values for the other mixtures are shown for comparison. The α values for any of the C_6 hydrocarbons are high in the region of lower x_2'' and decrease rapidly with an increase of x_2'' . At 298.15 K cyclohexane gives small values below 10 for α , but at 323.15 K has very large ones in the x_2'' range less than 0.1, comparable to those for benzene at 298.15 K; the values for *n*-hexane are in position between the two smooth

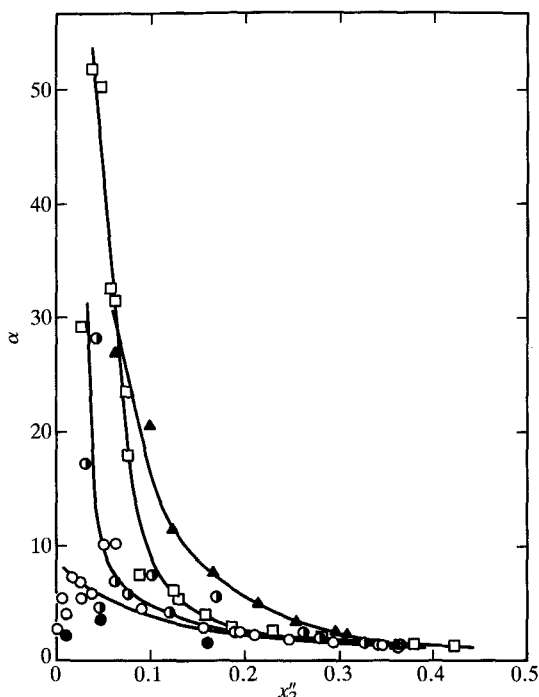


FIGURE 4. Separation factor α as a function of mole fraction x_2'' of C_2H_5OH in the organic phase. ○, and ● (Tarasenkov and Paul'sen),⁽⁵⁾ cyclohexane (298.15 K); □, cyclohexane (323.15 K); ●, *n*-hexane (298.15 K);⁽²⁾ ▲, benzene (298.15 K).⁽³⁾

curves drawn through the points for cyclohexane. From the observations for D_2 and α , it is concluded that at relatively high temperatures cyclohexane can be used as an effective extractant for ethanol, as well as benzene.

REFERENCES

1. Moriyoshi, T.; Sakamoto, T.; Uosaki, Y. *J. Chem. Thermodynamics* **1989**, 21, 947.
2. Moriyoshi, T.; Uosaki, Y.; Matsuura, H.; Nishimoto, W. *J. Chem. Thermodynamics* **1988**, 20, 551.
3. Yi-Chung, C.; Moulton, R. W. *Ind. Eng. Chem.* **1953**, 45, 2350.
4. Vold, R. D.; Washburn, E. R. *J. Am. Chem. Soc.* **1932**, 54, 4217.
5. Tarasenkov, D. N.; Paul'sen, Ya. A. *Zh. Org. Khim.* **1937**, 7, 2143.
6. Riddick, J. A.; Bunger, W. B.; Sakano, T. K. *Organic Solvents*. Fourth edition. Wiley-Interscience: New York. **1986**.
7. Barton, A. F. M. *CRC Handbook of Solubility Parameters and Other Cohesion Parameters*. CRC Press, Inc.: Boca Raton, Florida. **1983**, p. 142.
8. Moriyoshi, T.; Uosaki, Y.; Sakamoto, T. *J. Chem. Thermodynamics* **1988**, 20, 589.